数理方程

第8周
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ODE级数解 $-\frac{d}{dx}$

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\lambda\Theta=0 \qquad 0\le\theta\le\pi,\ \Theta(0),\Theta(\pi)<\infty$$

$$x=\cos\theta \Rightarrow \frac{d}{d\theta}=\frac{dx}{d\theta}\frac{d}{dx}=-\sin\theta\frac{d}{dx} \Rightarrow -\frac{d}{dx}=\frac{1}{\sin\theta}\frac{d}{d\theta}$$

$$\Rightarrow -\frac{d}{dx}\left[\sin^2\theta\left(-\frac{d\Theta}{dx}\right)\right]+\lambda\Theta=0 \Rightarrow \frac{d}{dx}\left[(1-x^2)\frac{d\Theta}{dx}\right]+\lambda\Theta=0$$

$\Theta\to y$

Legendre方程

$$\frac{d}{dx}\left[(1-x^2)\frac{dy}{dx}\right]+\lambda y=0,\quad -1<x<1$$

端点 $p(\pm1)=0$，SL理论奇异情形，加有界条件：$y(\pm1)<\infty$

$\frac{d}{dx}\left[p(x)\frac{dy}{dx}\right]+\cdots=0$，

$p(x)=1-x^2$

$y''(x)+p(x)y'(x)+q(x)y(x)=0$

根据 $p(x)$, $q(x)$ 在 x_0 附近的行为分类：

x_0 为常点：$p(x)$, $q(x)$ 在 x_0 点解析 $\Leftrightarrow p(x_0), q(x_0)<\infty$

$p(x)=\sum_{k=0}^{\infty} A_k(x-x_0)^k$

$q(x)=\sum_{k=0}^{\infty} B_k(x-x_0)^k$

x_0 为奇点：$p(x)$ 或 $q(x)$ 在 x_0 不解析

$x\to x_0$ 时，$p(x)$ 或 $q(x)\to\infty$

正则奇点：$x\to x_0$ 时，$(x-x_0)p(x)<\infty$，$(x-x_0)^2q(x)<\infty$

$(x-x_0)p(x)=\sum_{k=0}^{\infty} A_k(x-x_0)^k$，$(x-x_0)^2q(x)=\sum_{k=0}^{\infty} B_k(x-x_0)^k$

$p(x)$ 最多 $\sim(x-x_0)^{-1}$　　$q(x)$ 最多 $\sim(x-x_0)^{-2}$

非正则奇点：$x\to x_0$ 时，$(x-x_0)p(x)\to\infty$ 或 $(x-x_0)^2q(x)\to\infty$

欧拉方程 $x^2y''+\alpha xy'+\beta y=0 \Rightarrow y''+\frac{\alpha}{x}y'+\frac{\beta}{x^2}y=0$，$x=0$ 是正则奇点（除线除过）

$p\sim x^{-1}$　$q\sim x^{-2}$

常点　定理：$p(x)$、$q(x)$ 在 x_0 的邻域 $|x-x_0|<R$ 内解析，则 $y''(x)+p(x)y'(x)+q(x)=0$，$y(x_0)=C_0$，$y'(x_0)=C_1$ 在 $|x-x_0|<R$ 内有唯一解，且解在此区域解析

可展开为级数 $y(x)=\sum_{k=0}^{\infty} a_k(x-x_0)^k$

例：Legendre方程　　$-1<x<1$，$y(\pm1)<\infty$

$$(1-x^2)y''-2xy'+\lambda y=0$$

$$y''-\frac{2x}{1-x^2}y'+\frac{\lambda}{1-x^2}y=0$$

$x=0$为常点，$x=\pm1$是奇点

在$x=0$附近求级数解

代入

不要代入右边这个形式，否则需要 $\frac{1}{1-x^2}=1+x^2+x^4+\cdots$，较复杂

$$y(x)=\sum_{k=0}^{\infty}a_kx^k=a_0+a_1x+a_2x^2+\cdots$$

$$\Rightarrow y'=a_1+2a_2x+3a_3x^2+\cdots$$

$$\begin{cases} y''=2\cdot1a_2+3\cdot2a_3x+4\cdot3a_4x^2+\cdots+(k+1)(k+2)a_{k+2}x^k+\cdots \\ -x^2y''=\quad -2\cdot1a_2x^2-\cdots-k(k-1)a_kx^k+\cdots \\ -2xy'=\quad -2a_1x-2\cdot2a_2x^2-\cdots-2ka_kx^k+\cdots \\ \lambda y=\lambda a_0+\lambda a_1x+\lambda a_2x^2+\cdots+\lambda a_kx^k+\cdots \end{cases}$$

$$\downarrow \qquad \downarrow \qquad \downarrow$$

$$2a_2+\lambda a_0=0$$

$$6a_3+(\lambda-2)a_1=0$$

$$(k+2)(k+1)a_{k+2}+[\lambda-k(k+1)]a_k=0$$

$$k\geq0$$

$a_0\to a_2,a_4,a_6\cdots$

$a_1\to a_3,a_5,a_7\cdots$

递推关系：

$$a_{k+2}=\frac{k(k+1)-\lambda}{(k+2)(k+1)}a_k$$

$k=2n-2$

$$a_{2n}=\frac{(2n-1)(2n-2)-\lambda}{2n(2n-1)}a_{2n-2}=(\cdots)a_{2n-4}=\cdots=\prod_{j=1}^{n}\frac{(2j-1)(2j-2)-\lambda}{2j(2j-1)}a_0$$

$k=2n-1$

$$a_{2n+1}=\frac{2n(2n-1)-\lambda}{(2n+1)2n}a_{2n-1}=\cdots=\prod_{j=1}^{n}\frac{2j(2j-1)-\lambda}{(2j+1)(2j)}a_1$$

$$\Rightarrow y(x) = (a_0 + a_2x^2 + \cdots) + (a_1x + a_3x^3 + \cdots)$$
$$= a_0 y_0(x) + a_1 y_1(x)$$

a_0, a_1：任意常数

$$y_0(x) = 1 + \sum_{n=1}^{\infty} \prod_{j=1}^{n} \frac{(2j-1)(2j-2) - \lambda}{2j(2j-1)} x^{2n}$$

$$y_1(x) = x + \sum_{n=1}^{\infty} \prod_{j=1}^{n} \frac{2j(2j-1) - \lambda}{(2j+1)(2j)} x^{2n+1}$$

收敛区间 $a_0 y_0(x) = a_0 + a_2x^2 + a_4x^4 + \cdots$

d'Alembert判别法：级数 $u_0 + u_1 + u_2 + \cdots$

若 $\lim_{k\to\infty} \left|\frac{u_{k+1}}{u_k}\right| \to$ 极限值 $\begin{cases} >1 \Rightarrow \text{发散} \\ <1 \Rightarrow \text{收敛} \\ =1 \Rightarrow \text{无法判别} \end{cases}$

$$\Rightarrow \begin{cases} |x|<1 & \text{收敛} \\ |x|>1 & \text{发散} \\ |x|=1 & \text{无法判别} \end{cases}, \quad \left|\frac{a_{2n+2}x^{2n+2}}{a_{2n}x^{2n}}\right| = |x|^2 \frac{2n(2n+1)-\lambda}{(2n+2)(2n+1)} \xrightarrow{n\to\infty} |x|^2$$

$a_1 y_1(x)$ 结论类似

$x = \pm 1$？ $(1-x^2)y'' - 2xy' + \lambda y = 0$，$-1 < x < 1$，$y(\pm 1) < \infty$

高斯判别法：正项级数 $\sum_{k=0}^{\infty} u_k$，$u_k > 0$

对足够大的 k，$u_k > 0$ 即可

若 $\frac{u_k}{u_{k+1}} \xrightarrow{k\to\infty} 1 + \frac{\mu}{k} + \frac{\theta_k}{k^2}$ $(k\to\infty, |\theta_k| < \infty)$

$$\Rightarrow \begin{cases} \mu > 1 \Rightarrow \sum_{k=0}^{\infty} u_k \text{ 收敛} \\ \mu \le 1 \Rightarrow \sum_{k=0}^{\infty} u_k \text{ 发散} \end{cases}$$

调和级数 $\sum_{k=1}^{\infty} \frac{1}{k}$，$\frac{u_k}{u_{k+1}} = \frac{k+1}{k} = 1 + \frac{1}{k}$

$\mu = 1$ 发散

对 $y_0(x) = a_0 + a_2x^2 + \cdots$，$|x| = 1$

$$\frac{a_{2n}x^{2n}}{a_{2n+2}x^{2n+2}} = \frac{a_{2n}}{a_{2n+2}} = \frac{(2n+2)(2n+1)}{2n(2n+1)-\lambda} \xrightarrow{n\to\infty} \frac{(2n+2)(2n+1)}{2n(2n+1)} \frac{1}{1-\frac{\lambda}{2n(2n+1)}}$$

$$= \left(1+\frac{1}{n}\right)\left[1+\frac{\lambda}{2n(2n+1)}+\left(\frac{\lambda}{2n(2n+1)}\right)^2+\cdots\right] = 1+\frac{1}{n}+\frac{\theta_n}{n^2},\ |\theta_n|<\infty$$

$\Rightarrow$ 发散

出路：级数退化为多项式

$$a_{k+2} = \frac{k(k+1)-\lambda}{(k+2)(k+1)}a_k, \quad \lambda = l(l+1),\ l=0,1,2,\cdots$$

$$a_{l+2} = \frac{l(l+1)-l(l+1)}{(l+2)(l+1)}a_l = 0 \Rightarrow a_{l+4} = a_{l+6} = \cdots = 0$$

$\Rightarrow$
- l 为偶：$y_0(x) = a_0 + a_2x^2 + \cdots + a_lx^l$ 退化　取 $a_1 = a_3 = \cdots = 0$
- l 为奇：$y_1(x) = a_1x + a_3x^3 + \cdots + a_lx^l$ 退化　取 $a_0 = a_2 = \cdots = 0$

$$P_l(x) = \begin{cases} y_0(x)/y_0(1) & l=\text{偶},\ \text{偶函数}\ P_l(x)=P_l(-x) \\ y_1(x)/y_1(1) & l=\text{奇}\quad \text{奇函数}\ P_l(x)=-P_l(-x) \end{cases}$$

$P_l(x)$：Legendre 多项式　　归一化 $P_l(1)=1$

$\Rightarrow$ Legendre 方程在 $y(\pm1)$ 有界这一边界条件下的本征值为

$\lambda_l = l(l+1)$，本征函数为 $P_l(x)$，$l=0,1,2\cdots$

$P_0(x) = 1$,

$P_1(x) = x$

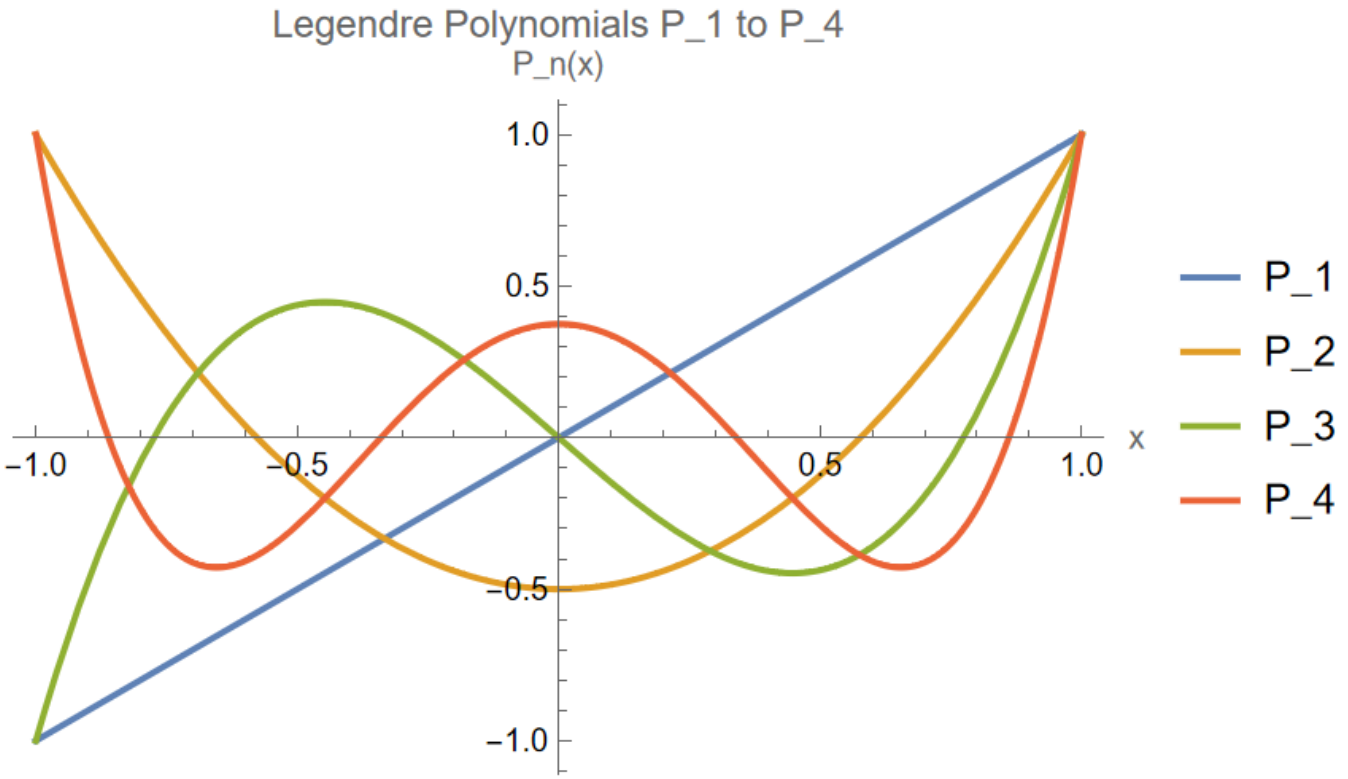
$P_2(x) = \frac{1}{2}(3x^2-1)$

$P_3(x) = \frac{1}{2}(5x^3-3x)$

$P_4(x) = \frac{1}{8}(35x^4-30x^2+3)$

$P_5(x) = \frac{1}{8}(63x^5-70x^3+15x)$

$$\int_{-1}^{1} P_l(x)P_k(x)dx = 0 \quad (l\neq k) \leftarrow \text{S-L 理论}$$


Legendre Polynomials P_1 to P_4
P_n(x)
1.0
0.5
−0.5
−1.0
−1.0
−0.5
0.5
1.0
x
P_1
P_2
P_3
P_4

正则奇点

例：Bessel方程 $x^2y''+xy'+(x^2-\nu^2)y=0$（$\nu$阶） $\Rightarrow y''+\frac{1}{x}y'+(1-\frac{\nu^2}{x^2})y=0$

$xp(x)<\infty$，$x^2q(x)<\infty$

$\Rightarrow x=0$为正则奇点.

Fuchs定理：在正则奇点x_0的去心邻域 $0<|x-x_0|<R$内，

有两个线性无关解：

$$y_1(x)=(x-x_0)^{\rho_1}\sum_{k=0}^{\infty}c_k(x-x_0)^k \quad c_0\neq 0$$

y_1, y_2 正则解

$$y_2(x)=g\,y_1(x)\ln(x-x_0)+(x-x_0)^{\rho_2}\sum_{k=0}^{\infty}d_k(x-x_0)^k,\ g\text{或}d_0\neq 0$$

广义幂级数解（Frobenius方法）

正则奇点幼儿园版

欧拉方程

$x^2y''+\alpha xy'+\beta y=0$（$\alpha,\beta$常数），$x=0$为正则奇点

$\Rightarrow y''+\frac{\alpha}{x}y'+\frac{\beta}{x^2}y=0$

$x\frac{d}{dx}(x\frac{dy}{dx})+(\alpha-1)x\frac{dy}{dx}+\beta y=0 \xrightarrow{x=e^t,\ x\frac{d}{dx}=\frac{d}{dt}} \frac{d^2y}{dt^2}+(\alpha-1)\frac{dy}{dt}+\beta y=0$

$y=e^{\rho t}\Rightarrow[\rho^2+(\alpha-1)\rho+\beta]e^{\rho t}=0\Rightarrow \rho^2+(\alpha-1)\rho+\beta=0$

$\Rightarrow \rho_{1,2}=-\frac{1}{2}[(\alpha-1)\pm\sqrt{(\alpha-1)^2-4\beta}]$ ← 不必为整数

$$\Rightarrow\begin{cases}\rho_1\neq\rho_2\Rightarrow y=c_1e^{\rho_1 t}+c_2e^{\rho_2 t}=c_1x^{\rho_1}+c_2x^{\rho_2}\\ \rho_1=\rho_2=\frac{1}{2}(1-\alpha)\Rightarrow y=(c_1+c_2t)e^{\rho_1 t}=c_1x^{\rho_1}+c_2x^{\rho_1}\ln x\end{cases}$$

当$(\alpha-1)^2-4\beta=0$

对数项！

对数项的另一种理解：$L[y]=x^2y''+\alpha xy'+\beta y=0$

$L=x^2\frac{d^2}{dx^2}+\alpha x\frac{d}{dx}+\beta$

$L[x^\rho]=\rho(\rho-1)x^\rho+\alpha\rho x^\rho+\beta x^\rho=[\rho^2+(\alpha-1)\rho+\beta]x^\rho=(\rho-\rho_1)(\rho-\rho_2)x^\rho$

$\rho_1\neq\rho_2$, $L[x^{\rho_1}]=L[x^{\rho_2}]=0\Rightarrow x^{\rho_1}, x^{\rho_2}$是解 $\Rightarrow y=c_1x^{\rho_1}+c_2x^{\rho_2}$

$\rho_1=\rho_2\Rightarrow L[x^{\rho_1}]=(\rho-\rho_1)^2x^\rho\Rightarrow$只得到一解$x^{\rho_1}$

得到第二解的巧妙手段：对ρ求导.

$$L[x^\rho]=(\rho-\rho_1)^2x^\rho$$

$$\Downarrow\frac{\partial}{\partial\rho}$$

$$\frac{\partial}{\partial\rho}L[x^\rho]=L[\frac{\partial}{\partial\rho}x^\rho]=L[\frac{\partial}{\partial\rho}e^{\rho\ln x}]=L[\ln x\cdot x^\rho]$$

$$=2(\rho-\rho_1)x^\rho+(\rho-\rho_1)^2x^\rho\ln x\xrightarrow{\rho=\rho_1}0$$

$$\Rightarrow L[\ln x\cdot x^\rho]\Big|_{\rho=\rho_1}=0$$

即 $x^{\rho_1}\ln x$ 是欧拉方程的解.

欧拉方程 $\begin{cases}\alpha\longrightarrow A_0+A_1x+A_1x^2+\cdots\\ \beta\longrightarrow B_0+B_1x+B_2x^2+\cdots\end{cases}$ $\Rightarrow$ 一般的正则奇点.

一般正则奇点

$x=0$ 为正则奇点 $\Rightarrow$

$y''+p(x)y'+q(x)y=0$, $xp(x)=A_0+A_1x+A_2x^2+\cdots$, $x^2q(x)=B_0+B_1x+\cdots$

$\Rightarrow x^2y''+x[xp(x)]y'+[x^2q(x)]y=0$

$\Rightarrow x^2y''+x(A_0+A_1x+\cdots)y'+(B_0+B_1x+\cdots)y=0$ ← 只保留 A_0, B_0，则回到欧拉方程.

$y=\sum_{k=0}^{\infty}a_kx^{k+\rho}$ $(a_0\neq 0)$, $y'=\sum_{k=0}^{\infty}a_k(k+\rho)x^{k+\rho-1}$

↑ 由欧拉方程启发

$y''=\sum_{k=0}^{\infty}a_k(k+\rho)(k+\rho-1)x^{k+\rho-2}$

$L[y]=x^2y''+x(A_0+A_1x+\cdots)y'+(B_0+B_1x+\cdots)y$, $L[y]=0$

$\Rightarrow L[y]=x^{\rho}\Big[\sum_{k=0}^{\infty}(k+\rho)(k+\rho-1)a_kx^k+\Big(\sum_{m=0}^{\infty}A_mx^m\Big)\Big(\sum_{k=0}^{\infty}(k+\rho)a_kx^k\Big)+\Big(\sum_{m=0}^{\infty}B_mx^m\Big)\Big(\sum_{k=0}^{\infty}a_kx^k\Big)\Big]$

$=[\rho(\rho-1)+A_0\rho+B_0]a_0x^{\rho}+\sum_{k=1}^{\infty}\Big\{[(k+\rho)(k+\rho-1)+A_0(k+\rho)+B_0]a_k+\sum_{m=0}^{k-1}[(m+\rho)A_{k-m}+B_{k-m}]a_m\Big\}x^{k+\rho}$

$=f_0(\rho)$

记为 F_k. 要求 $F_k=0$ $(k\geq 1)$

$\Rightarrow f_0(k+\rho)a_k=-\sum_{m=0}^{k-1}[(m+\rho)A_{k-m}+B_{k-m}]a_m$

$\rho\to k+\rho$

$a_k=-\frac{1}{f_0(k+\rho)}\sum_{m=0}^{k-1}[(m+\rho)A_{k-m}+B_{k-m}]a_m$

$=a_0f_0(\rho)x^{\rho}+\sum_{k=1}^{\infty}F_kx^{k+\rho}$

递推关系：$a_k=(\cdots)a_{k-1}+(\cdots)a_{k-2}+\cdots+(\cdots)a_0$

ρ 满足指标方程 $f_0(\rho)=0$

$\{a_k\}$ 满足递推关系 $F_k=0$ $(k\geq 1)$

$\Rightarrow L[y]=0$

广义幂级数解 $y=\sum_{k=0}^{\infty}a_kx^{k+\rho}$

a_0 为任意常数，$a_1, a_2, a_3\ldots$ 由递推关系决定，依赖 ρ，且正比于 a_0

$f_0(\rho) = \rho(\rho-1) + A_0\rho + B_0 = (\rho-\rho_1)(\rho-\rho_2)$

(1) $\rho_1 - \rho_2 \neq$ 整数 $\Rightarrow y_1 = \sum_{k=0}^{\infty} a_k(\rho_1) x^{k+\rho_1}$, $y_2 = \sum_{k=0}^{\infty} a_k(\rho_2) x^{k+\rho_2}$

(2) $\rho_1 = \rho_2$, 仅能得一解 $y = \sum_{k=0}^{\infty} a_k(\rho_1) x^{k+\rho_1}$

(另一解见下文)

(3) $\rho_1 - \rho_2 = n$ (正整数) 第一解 $y = \sum_{k=0}^{\infty} a_k(\rho_1) x^{k+\rho_1}$

表面看来和情况(1)并无不同, 实则不然.

当求 $y_2 = \sum_{k=0}^{\infty} a_k x^{k+\rho_2}$ ($a_0 \neq 0$) 时, 递推关系 $F_n = 0$ 出现意外:

$f_0(\rho_2+n)a_n = (\cdots)a_{n-1} + (\cdots)a_{n-2} + \cdots + (\cdots)a_0$

$= (\cdots)a_0$

由于递推关系, $a_{n-1}, a_{n-2} \cdots, a_1$ 正比于 a_0

$\rho_1 = \rho_2 + n$ 是指标方程之根, 即 $f_0(\rho_2+n) = 0$

$\Rightarrow 0 \cdot a_n = (\cdots)a_0$

$\frac{1}{2}$阶与1阶Bessel方程分别属于情况(a)(b), 见作业

$\Rightarrow$

(a) a_0的系数 $= 0 \Rightarrow a_n$任意 $\Rightarrow y_2(x)$含 a_0, a_n两个任意常数, 是方程通解. (取 $a_0 = 0, a_n \neq 0$ 可得 $y_2 = \sum_{k=n}^{\infty} a_k x^{k+\rho_2}$, 此级数的首项是 $x^{n+\rho_2} = x^{\rho_1}$, 所以级数其实正比于 $y_1 = \sum_{k=0}^{\infty} a_k x^{k+\rho_1}$. 因此通解包含了 y_1 作为特例. 取 $a_0 \neq 0, a_n = 0$ 即可得与 y_1 独立的第二解.)

(b) a_0的系数 $\neq 0 \Rightarrow$ 递推关系无法满足 $\Rightarrow$ 无 $y_2 = \sum_{k=0}^{\infty} a_k x^{k+\rho_2}$ 形式的解 $\Rightarrow$ 第二解要另行求出. (含对数项)

若取 $a_0 = \cdots = a_{n-1} = 0, a_n \neq 0, a_{n+1}, a_{n+2} \cdots$ 由递推得到, 则此解回到 $y_1 = \sum_{k=0}^{\infty} a_k x^{k+\rho_1}$

第二解含对数项时的求法：

(A) 对于情况(2)，$\rho_1=\rho_2$，第二解可由对ρ求导得出。

取 $y(x;\rho)=\sum_{k=0}^{\infty} a_k(\rho)x^{k+\rho}$ 满足递推关系 $F_k=0\ (k\geq 1)$，

a_0为常数，$a_1,a_2,a_3\cdots$是ρ的函数。暂不要求ρ满足指标方程。

$$\Rightarrow L[y(x;\rho)]=a_0 f_0(\rho)x^{\rho}=a_0(\rho-\rho_1)(\rho-\rho_2)x^{\rho}\underset{\rho_1=\rho_2}{=\!=}a_0(\rho-\rho_1)^2x^{\rho}$$

$\Downarrow \frac{\partial}{\partial\rho}$

直观理解：

$\rho_1\neq\rho_2$时$y(x;\rho_1),y(x;\rho_2)$都是解，二者之差也是解；$\rho_2\to\rho_1$时，两解之差化为导数 $\left.\frac{\partial y}{\partial\rho}\right|_{\rho=\rho_1}$

$$\frac{\partial}{\partial\rho}L[y(x;\rho)]=L\left[\frac{\partial y}{\partial\rho}\right]=\frac{\partial}{\partial\rho}\left[a_0(\rho-\rho_1)^2x^{\rho}\right]$$

$$=2a_0(\rho-\rho_1)x^{\rho}+a_0(\rho-\rho_1)^2x^{\rho}\ln x\ \underset{\rho=\rho_1}{=\!=}\ 0$$

$$\Rightarrow L\left[\left.\frac{\partial y}{\partial\rho}\right|_{\rho=\rho_1}\right]=0 \quad \text{即}\ \left.\frac{\partial y(x;\rho)}{\partial\rho}\right|_{\rho=\rho_1}\ \text{是第二解.}$$

$$y=\sum_{k=0}^{\infty}a_k(\rho)x^{k+\rho}\Rightarrow\left.\frac{\partial y}{\partial\rho}\right|_{\rho=\rho_1}=\sum_{k=0}^{\infty}\left(a_k x^{k+\rho}\ln x+\frac{\partial a_k}{\partial\rho}x^{k+\rho}\right)_{\rho=\rho_1}$$

$$=y_1(x)\ln x+\sum_{k=0}^{\infty}\left(\frac{\partial a_k}{\partial\rho}\right)_{\rho=\rho_1}x^{k+\rho_1}\ \equiv\ y_2(x)$$

$$\uparrow\quad y_1(x)=\sum_{k=0}^{\infty}a_k(\rho_1)x^{k+\rho_1}$$

以上对ρ求导的Frobenius方法不仅解释了对数项的来源，也给出了第二解的计算方法。（下文有计算实例）

另一种求解$y_2(x)$的方法是将 $y_2(x)=y_1(x)\ln x+\sum_{k=0}^{\infty}d_k x^{k+\rho_1}$ 形式代入微分方程，根据递推关系，求出系数d_k. 计算通常比较复杂。

作要求

(B) 对于情况(3)(b)，即 $\rho_1-\rho_2=n$（正整数），且第二解无法写成 $y_2=\sum_{k=0}^{\infty}a_k x^{k+\rho_2}$ 形式时，仍可通过对 ρ 求导，但方式有所不同。若照搬 $\rho_1=\rho_2$ 时的方式：

$$L[y(x;\rho)]=a_0(\rho-\rho_1)(\rho-\rho_2)x^{\rho}$$

$$\Downarrow \frac{\partial}{\partial\rho}$$

$$L\left[\frac{\partial y}{\partial\rho}\right]=a_0(2\rho-\rho_1-\rho_2)x^{\rho}+a_0(\rho-\rho_1)(\rho-\rho_2)x^{\rho}\ln x$$

$\neq 0$，无论 ρ 取何值

关键技巧是取 $a_0=\tilde{a}_0(\rho-\rho_2)$（$\tilde{a}_0$ 常数）$\Rightarrow a_1|_{\rho=\rho_2}=\cdots=a_{n-1}|_{\rho=\rho_2}=0$，$a_n|_{\rho=\rho_2}$ 任意

$$\Rightarrow \frac{\partial}{\partial\rho}L[y(x;\rho)]=L\left[\frac{\partial y}{\partial\rho}\right]=\frac{\partial}{\partial\rho}\left[\tilde{a}_0(\rho-\rho_1)(\rho-\rho_2)^2x^{\rho}\right]$$

$$=\tilde{a}_0(\rho-\rho_2)^2x^{\rho}+2\tilde{a}_0(\rho-\rho_1)(\rho-\rho_2)x^{\rho}+\tilde{a}_0(\rho-\rho_1)(\rho-\rho_2)^2x^{\rho}\ln x$$

$\rho=\rho_2$ 0

$$\Rightarrow L\left[\frac{\partial y}{\partial\rho}\Big|_{\rho=\rho_2}\right]=0,\ \text{即}\ \frac{\partial y}{\partial\rho}\Big|_{\rho=\rho_2}\ \text{是第二解}$$

- $\rho_1-\rho_2\neq$ 整数，$g=0$（无对数项）
- $\rho_1=\rho_2$，$g\neq 0$
- $\rho_1-\rho_2=$ 正整数，$g\neq 0$，$g=0$ 均有可能

作业题之一说明何时 $\begin{cases}g\neq 0\\ g=0\end{cases}$

$\rho_1=\rho_2$ 例

例：在 $x=0$ 附近求解 $xy''+y'-4y=0$

$y=\sum_{k=0}^{\infty}a_k x^{k+\rho}$ 代入 $x^2y''+xy'-4xy=0$ ← 试试把 $4x$ 换成 4 ⇒ 级数解什么形式？

$$x^2y''=\rho(\rho-1)a_0x^{\rho}+(\rho+1)\rho a_1x^{\rho+1}+\cdots+(\rho+k)(\rho+k-1)a_kx^{\rho+k}+\cdots$$
$$xy'=\rho a_0x^{\rho}+(\rho+1)a_1x^{\rho+1}+\cdots+(\rho+k)a_kx^{\rho+k}+\cdots$$
$$-4xy=-4a_0x^{\rho+1}-\cdots-4a_{k-1}x^{\rho+k}-\cdots$$

指标方程 $f_0(\rho)=\rho(\rho-1)+\rho=\rho^2\Rightarrow\rho_1=\rho_2=0$

递推关系 $F_k=0\Rightarrow a_k=\dfrac{4}{(\rho+k)^2}a_{k-1}=\cdots=\dfrac{4^k}{(\rho+1)^2\cdots(\rho+k)^2}a_0$

取 $\rho=\rho_1=0$

$$\Longrightarrow y_1(x)=1+\sum_{k=1}^{\infty}\frac{4^k}{(k!)^2}x^k\quad(a_0=1)$$

$$y_2=\left.\frac{\partial y(x;\rho)}{\partial\rho}\right|_{\rho=\rho_1}=\left.\frac{\partial}{\partial\rho}\left(\sum_{k=0}^{\infty}a_kx^{k+\rho}\right)\right|_{\rho=\rho_1}$$

$$=\sum_{k=0}^{\infty}\left(a_kx^{k+\rho}\ln x+\frac{\partial a_k}{\partial\rho}x^{k+\rho}\right)_{\rho=\rho_1}$$

取 $a_0=1$

$$=y_1(x)\ln x-2\sum_{k=1}^{\infty}\frac{4^k}{(\rho+1)^2\cdots(\rho+k)^2}\left(\frac{1}{\rho+1}+\cdots+\frac{1}{\rho+k}\right)x^{\rho+k}\Bigg|_{\rho=\rho_1=0}$$

$$=y_1(x)\ln x-2\sum_{k=1}^{\infty}\frac{4^kH_k}{(k!)^2}x^k$$

$H_k=1+\frac{1}{2}+\cdots+\frac{1}{k}$

例：Bessel方程 $x^2y''+xy'+(x^2-\nu^2)y=0$ $\nu\neq$整数或半整数

（去掉x^2 ⇒ 欧拉方程）

$x=0$是正则奇点

（见作业）

$$y(x)=\sum_{k=0}^{\infty}a_k x^{\rho+k}=a_0x^{\rho}+a_1x^{\rho+1}+\cdots \quad (a_0\neq 0)$$

$$x^2y''+xy'+(x^2-\nu^2)y=a_0f_0(\rho)x^{\rho}+[(\rho+1)^2-\nu^2]a_1x^{\rho+1}+\sum_{k=2}^{\infty}\{[(\rho+k)^2-\nu^2]a_k+a_{k-2}\}x^{\rho+k}$$

$f_0(\rho)=\rho^2-\nu^2=0$

$\Rightarrow \rho_1=\nu,\ \rho_2=-\nu$

先考虑 $\rho=\rho_1=\nu$，$a_1[(\rho+1)^2-\nu^2]=(\rho+1+\nu)(\rho+1-\nu)a_1=0$

$=(2\nu+1)\cdot 1\cdot a_1\neq 0$

$\Rightarrow a_1=0$

$$a_k=-\frac{1}{(\rho+k)^2-\nu^2}a_{k-2}\quad (k\geq 2)$$

$$=-\frac{1}{(\rho+k+\nu)(\rho+k-\nu)}a_{k-2}=-\frac{1}{k(2\nu+k)}a_{k-2}$$

$$a_{2n}=-\frac{1}{2n(2\nu+2n)}a_{2n-2}=-\frac{1}{4}\frac{1}{n(\nu+n)}a_{2n-2}$$

$$=(-1)^n\frac{1}{n!(\nu+n)(\nu+n-1)\cdots(\nu+1)}\frac{1}{2^{2n}}a_0$$

$$\Rightarrow y_1(x)=a_0x^{\nu}\left[1-\frac{1}{\nu+1}\left(\frac{x}{2}\right)^2+\frac{1}{2!}\frac{1}{(\nu+1)(\nu+2)}\left(\frac{x}{2}\right)^4+\cdots\right.$$

$$\left.(-1)^n\frac{1}{n!}\frac{1}{(\nu+1)\cdots(\nu+n)}\left(\frac{x}{2}\right)^{2n}+\cdots\right]$$

$$\lim_{k\to\infty}\frac{a_k x^k}{a_{k-2}x^{k-2}} = x^2 \lim_{k\to\infty}\left[-\frac{1}{k(2\nu+k)}\right] = 0$$

收敛半径 $R=\infty$,

$\Gamma(x+1)=\int_0^\infty t^x d(-e^{-t}) = -t^x e^{-t}\Big|_0^\infty + \int_0^\infty x t^{x-1} e^{-t}dt = x\Gamma(x)$

习惯取 $a_0 = \frac{1}{2^\nu \Gamma(\nu+1)}$ $\quad \Gamma(x)=\int_0^\infty e^{-t}t^{x-1}dt \quad (x>0)$

对 $x<0$, $\Gamma(x)$ 由 $\Gamma(x+1)=x\Gamma(x)$ 定义

$$\Downarrow$$

$$a_{2n} = (-1)^n \frac{1}{n!(\nu+n)(\nu+n-1)\cdots(\nu+1)} \frac{1}{2^{2n}} \frac{1}{2^\nu\Gamma(\nu+1)} a_0$$

$$= (-1)^n \frac{1}{2^{2n+\nu}} \frac{1}{n!(\nu+n)\cdots(\nu+2)\Gamma(\nu+2)}$$

$\Gamma(\nu+2)=(\nu+1)\Gamma(\nu+1)$

$= \cdots$ (继续应用 $x\Gamma(x)=\Gamma(x+1)$)

$$= (-1)^n \frac{1}{2^{2n+\nu}} \frac{1}{n!\Gamma(\nu+n+1)}$$

$$\Rightarrow y_1(x) = \sum_{n=0}^\infty (-1)^n \frac{1}{n!\Gamma(\nu+n+1)} \left(\frac{x}{2}\right)^{2n+\nu} = J_\nu(x)$$

ν 阶 Bessel 函数

同理，取 $\rho=\rho_2=-\nu$,

$$\Rightarrow y_2(x) = b_0 x^{-\nu}\left[1 - \frac{1}{-\nu+1}\left(\frac{x}{2}\right)^2 + \frac{1}{2!(-\nu+1)(-\nu+2)}\left(\frac{x}{2}\right)^4 + \cdots + (-1)^n \frac{1}{n!(-\nu+1)(-\nu+2)\cdots(-\nu+n)}\left(\frac{x}{2}\right)^{2n} + \cdots\right]$$

取 $b_0 = \frac{1}{2^{-\nu}\Gamma(-\nu+1)} = \frac{2^\nu}{\Gamma(-\nu+1)}$

$$\Rightarrow y_2(x) = \sum_{n=0}^\infty \frac{(-1)^n}{n!\Gamma(-\nu+n+1)}\left(\frac{x}{2}\right)^{2n-\nu} = J_{-\nu}(x)$$

$-\nu$ 阶 Bessel 函数.

通解 $y(x) = C_1 J_\nu(x) + C_2 J_{-\nu}(x)$, $\quad 0 < |x| < \infty$

非正则奇点?

例: $y'' - \frac{5}{x^3} y = 0$, $\quad x=0$ 非正则, $-\frac{5}{x^3} x^2 \to \infty$

若设 $y(x) = \sum_{k=0}^{\infty} a_k x^{\rho+k}$ $(a_0 \neq 0)$, 代入ODE

$\Rightarrow$ 仅有的 $x^{\rho-3}$ 项为 $-5a_0 x^{\rho-3} = 0$ $\Rightarrow a_0 \neq 0$, 与 $a_0 \neq 0$ 矛盾.

$\Rightarrow$ 无 $y(x) = \sum_{k=0}^{\infty} a_k x^{\rho+k}$ 形式之解!